LAPORAN TUGAS PEKAN 6

STRUKTUR DATA

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```java
package pekan6_2511531015;
import java.util.Scanner;

public class Musik_2511531015 {

    // Nama kelas mengikuti aturan: Musik_2511531015

        private Lagu_2511531015 head_1015;

        // Constructor
        public Musik_2511531015() {
            head_1015 = null;
        }

        // Menambah lagu di akhir playlist
        public void tambahLagu_1015(String judul, String penyanyi) {
            Lagu_2511531015 newLagu = new Lagu_2511531015(judul, penyanyi);
            if (head_1015 == null) {
                head_1015 = newLagu;
            } else {
                Lagu_2511531015 temp = head_1015;
                while (temp.next_1015 != null) {
                    temp = temp.next_1015;
                }
                temp.next_1015 = newLagu;
                newLagu.prev_1015 = temp;
            }
            System.out.println("Lagu berhasil ditambahkan!");
        }

        // Menghapus lagu pertama (head)
        public void hapusLaguAwal_1015() {
            if (head_1015 != null) {
                head_1015 = head_1015.next_1015;
                if (head_1015 != null) {
                    head_1015.prev_1015 = null;
                }
                System.out.println("Lagu pertama berhasil dihapus!");
            } else {
                System.out.println("Playlist kosong!");
            }
        }

        // Menampilkan playlist dari awal ke akhir
        public void tampilMaju_1015() {
            if (head_1015 == null) {
                System.out.println("Playlist kosong!");
            } else {
```

```java
            if (!found) {
                System.out.println("Lagu tidak ditemukan!");
            }
        }

        public static void main(String[] args) {
            Scanner scanner = new Scanner(System.in);
            Musik_2511531015 playlist = new Musik_2511531015();

            while (true) {
                System.out.println("\n=== Playlist Musik NIM: 2511531015 ===");
                System.out.println("1. Tambah Lagu");
                System.out.println("2. Hapus Lagu Pertama");
                System.out.println("3. Lihat Playlist (Maju)");
                System.out.println("4. Lihat Playlist (Mundur)");
                System.out.println("5. Cari Lagu");
                System.out.println("6. Keluar");
                System.out.print("Pilihan: ");
                int pilihan = scanner.nextInt();
                scanner.nextLine();  // Consume newline

                if (pilihan == 1) {
                    System.out.print("Judul: ");
                    String judul = scanner.nextLine();
                    System.out.print("Penyanyi: ");
                    String penyanyi = scanner.nextLine();
                    playlist.tambahLagu_1015(judul, penyanyi);
                } else if (pilihan == 2) {
                    playlist.hapusLaguAwal_1015();
                } else if (pilihan == 3) {
                    playlist.tampilMaju_1015();
                } else if (pilihan == 4) {
                    playlist.tampilMundur_1015();
                } else if (pilihan == 5) {
                    System.out.print("Masukkan judul lagu yang dicari: ");
                    String judul = scanner.nextLine();
                    playlist.cariLagu_1015(judul);
                } else if (pilihan == 6) {
                    break;
                } else {
                    System.out.println("Pilihan tidak valid!");
                }
            }
            scanner.close();
        }
    }
```

1. Class Definition:

- The class Musik_2511531015 manages a playlist of songs.

- It has a private member variable head_1015 which points to the first song in the playlist (Lagu_2511531015 is a song object).
- The class provides various methods for interacting with the playlist, such as adding a song, deleting the first song, and displaying the playlist.

2. Methods:

- **Constructor:**
  - The constructor Musik_2511531015() initializes the playlist by setting the head_1015 (the first song) to null, meaning that initially the playlist is empty.
- **tambahLagu_1015(String judul, String penyanyi):**
  - This method adds a new song to the end of the playlist.
  - It creates a new song object (Lagu_2511531015), sets its title (judul) and artist (penyanyi), and appends it to the end of the list.
  - If the playlist is empty, it makes the new song the first song in the list.
- **hapusLaguAwal_1015():**
  - This method deletes the first song in the playlist (the "head" of the linked list).
  - If there is only one song, it sets head_1015 to null (i.e., the playlist becomes empty).
  - If the playlist has more than one song, it adjusts the head_1015 pointer to the next song and sets the new head's prev_1015 to null.
- **tampilMaju_1015():**
  - This method prints the songs in the playlist from the first to the last song.
  - It traverses the linked list starting from head_1015 and prints the title and artist of each song.
- **tampilMundur_1015():**
  - This method prints the songs in the playlist in reverse order (from the last song to the first).
  - It first finds the last song by traversing to the end of the list, and then it prints the songs while traversing backward.
- **cariLagu_1015(String judul):**
  - This method searches for a song by its title (judul).

    - It compares the song titles in a case-insensitive manner and prints the song's details if found.

3. Main Program (main method):

- The main method provides an interactive menu for the user to manage the playlist.
- The menu allows the user to:
    - Add a song to the playlist.
    - Remove the first song.
    - View the playlist from the start to the end (forward).
    - View the playlist from the end to the start (reverse).
    - Search for a song by its title.
- The program loops until the user selects the option to exit.

4. Lagu_2511531015 Class:

- This class is used to represent individual songs in the playlist.
- It has properties such as judul (title) and penyanyi (artist), along with pointers (next_1015 and prev_1015) to the next and previous songs in the list.
- Methods like getJudul_1015() and getPenyanyi_1015() are assumed to return the title and artist of the song, respectively.

5. User Interaction:

- The program uses a Scanner to get input from the user.
- Depending on the user's choice, the program calls the corresponding method to perform operations on the playlist.

Example of Flow:

- When the user selects **1** to add a song, they are prompted to enter the title and artist of the song. The song is then added to the playlist.
- If the user selects **2** to remove the first song, the first song is removed from the playlist.
- If the user selects **3**, the playlist is displayed from the first to the last song.
- If the user selects **4**, the playlist is displayed in reverse order.
- If the user selects **5**, they can search for a song by title.

```java
package pekan6_2511531015;

public class Lagu_2511531015 {
    // Nama kelas mengikuti aturan: Lagu_2511531015
        private String judul_1015;
        private String penyanyi_1015;
        public Lagu_2511531015 next_1015;
        public Lagu_2511531015 prev_1015;

        // Constructor
        public Lagu_2511531015(String judul, String penyanyi) {
            this.judul_1015 = judul;
            this.penyanyi_1015 = penyanyi;
            this.next_1015 = null;
            this.prev_1015 = null;
        }

        // Getter dan Setter
        public String getJudul_1015() {
            return judul_1015;
        }

        public void setJudul_1015(String judul) {
            this.judul_1015 = judul;
        }

        public String getPenyanyi_1015() {
            return penyanyi_1015;
        }

        public void setPenyanyi_1015(String penyanyi) {
            this.penyanyi_1015 = penyanyi;
        }
    }
```

1. Class Definition:

- **Class Name:** Lagu_2511531015
    - This class is used to represent a song, following a naming convention that includes the user's NIM (2511531015).
- The class has two main attributes:
    - judul_1015 (title of the song).
    - penyanyi_1015 (artist of the song).
    - Two pointers (next_1015 and prev_1015) are used to create a doubly linked list structure for the songs in the playlist.

2. Instance Variables (Fields):

- **private String judul_1015;**
    - This is a private instance variable that stores the title of the song.
- **private String penyanyi_1015;**
    - This is a private instance variable that stores the name of the artist or singer of the song.
- **public Lagu_2511531015 next_1015;**
    - This public instance variable is a reference (pointer) to the next song in the playlist (used for the linked list structure).

- **public Lagu_2511531015 prev_1015;**
    - This public instance variable is a reference (pointer) to the previous song in the playlist (also for the linked list structure).

3. Constructor:

- **public Lagu_2511531015(String judul, String penyanyi)**
    - This is the constructor of the Lagu_2511531015 class, which is used to initialize the song object with a given title and artist.
    - The constructor accepts two parameters: judul (song title) and penyanyi (song artist).
    - Inside the constructor:
        - The title (judul_1015) and artist (penyanyi_1015) are set to the values passed in as parameters.
        - Both the next_1015 and prev_1015 pointers are set to null initially. These pointers are used to link the song to other songs in the playlist.

4. Getter and Setter Methods:

- **public String getJudul_1015()**
    - This is a getter method that returns the title of the song (judul_1015).
- **public void setJudul_1015(String judul)**
    - This is a setter method that allows the song's title (judul_1015) to be modified.
- **public String getPenyanyi_1015()**
    - This is a getter method that returns the artist of the song (penyanyi_1015).
- **public void setPenyanyi_1015(String penyanyi)**
    - This is a setter method that allows the song's artist (penyanyi_1015) to be modified.

5. Purpose of the Class:

- The class Lagu_2511531015 represents an individual song in the playlist system.
- It contains attributes for the song's title and artist, and uses the next_1015 and prev_1015 pointers to form a doubly linked list structure.
- The class provides getter and setter methods to access and modify the title and artist of the song.

- By using the next_1015 and prev_1015 pointers, the class enables navigation between songs in both directions (forward and backward) in the playlist.

6. Usage:

- This class is used in conjunction with the Musik_2511531015 class, which manages the playlist as a whole.
- Each Lagu_2511531015 object represents a single song, and the playlist is built by linking these song objects together using the next_1015 and prev_1015 pointers.
- The playlist can be navigated forward and backward using these pointers in the Musik_2511531015 class.

KESIMPULAN

Kode program ini terdiri dari dua kelas utama: Musik_2511531015 dan Lagu_2511531015. Kelas Lagu_2511531015 berfungsi untuk merepresentasikan setiap lagu dalam playlist dengan menyimpan informasi tentang judul lagu dan penyanyi, serta menghubungkan lagu-lagu dalam playlist menggunakan dua pointer (next_1015 dan prev_1015) yang membentuk struktur linked list ganda.

Sementara itu, kelas Musik_2511531015 berfungsi untuk mengelola playlist secara keseluruhan. Kelas ini menyediakan berbagai metode untuk menambah, menghapus, menampilkan, dan mencari lagu di dalam playlist. Playlist ini menggunakan kelas Lagu_2511531015 untuk setiap elemen lagu, yang dapat ditelusuri maju maupun mundur dengan memanfaatkan pointer-pointer yang ada.

Secara keseluruhan, program ini merupakan implementasi sederhana dari sistem playlist musik menggunakan linked list ganda di mana pengguna dapat mengelola lagu dalam playlist dengan menambah, menghapus, dan menampilkan lagu-lagu berdasarkan pilihan yang ada.